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Published: 07 July 2001

[Citation in BibTeX format](#)

The Mixture of Trees Factorized Distribution Algorithm

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Abstract

This paper introduces a Factorized Distribution Algorithm based on a mixture of trees distribution. The probabilistic model and the learning algorithm used differs to previous uses of probabilistic modeling in the context of Evolutionary Computation. Preliminary results show the algorithm is competitive, and some times superior to other Factorized Distribution Algorithms. We also illustrate how particular features of the search space can be employed during the search by conveniently selecting the mixture of trees parameters.

1 INTRODUCTION

Factorized Distribution Algorithms (FDAs) (Mühlenbein, Mahnig, & Ochoa, 1999) are population based search methods that combine results from Graphical Models and Evolutionary Computation research, and are considered as a tractable subclass of Estimation Distribution Algorithms (Mühlenbein & Paaß, 1996). In order to optimize a given function they begin by generating an initial random population of points which are evaluated using the objective function. Some of the points are selected based on their values, and a factorized probabilistic model of their underlying distribution is constructed. This probabilistic model is used to sample the points that will be part of the next population.

A number of FDAs that use probabilistic models based on dependency trees have shown up in the literature. In (Baluja & Davies, 1997) a tree factorization computed using the Chow and Liu algorithm (Chow &

Liu, 1968) is employed to approximate the underlying distribution of the selected points. The Bivariate Marginal Distribution Algorithm (BMDA) (Pelikan & Mühlenbein, 1999) uses a forest instead of a tree based factorization. Only second order statistics are computed.

The Polytree Approximation Distribution Algorithm (PADA) has been designed to deal with the class of single connected Bayesian Networks (BNs) (Soto *et al.*, 1999). A revised version of this algorithm has been presented in (Ochoa, Muehlenbein, & Soto, 2000) where the previous introduced algorithms based on trees and forest distributions are covered.

Modeling by finite mixture of distributions (Everitt & Hand, 1981) concerns modeling a statistical distribution by a mixture (or weighted sum) of other distributions. Recently, the research on mixture models has begun to receive a particular attention by the EDAs community. In (Thierens & Bosman, 2001) the authors use a mixture of Gaussian probabilistic density functions for the solution of continuous multi-objective functions. In (Peña, Lozano, & Larrañaga, 2001) authors employ mixture models as the basis for data clustering in multimodal continuous and discrete function optimization via EDAs. In the case of discrete optimization a framework for learning mixture of BNs that share the same structure is presented.

Mixture of distributions have been used also in the framework of evolutionary optimization that use flexible probability estimators (Gallagher, Frea, & Downs, 1999) (an approach very close to FDAs) where the adaptive mixture model of Priebe is used to estimate probability distributions in an evolutionary optimization context.

2 TREES AND MIXTURE OF TREES MODELS

Mixtures of trees belong to the class of finite mixture distributions. They were introduced in (Meila, 1999) and are the core of the optimization algorithm we present in this paper. Now the probabilistic models are formally introduced. We will utilize the same notation used in (Meila, 1999).

Let V denote the set of variables of our problem. According to the graphical model paradigm, each variable is viewed as a vertex of an (undirected) graph $G = (V, E)$ which is called a tree if it has no cycles. Now we define a probability distribution T that is conformal with a tree.

$$T(x) = \prod_{v \in V} T_{v|pa(v)}(x_v | x_{pa(v)}) \quad (1)$$

The distribution T itself will be called a tree when no confusion is possible. The graph (V, E) represents the structure of the distribution T .

A mixture of trees is defined to be a distribution of the form:

$$Q(x) = \sum_{k=1}^m \lambda_k T^k(x) \quad (2)$$

with $\lambda_k \geq 0$, $k = 1, \dots, m$, $\sum_{k=1}^m \lambda_k = 1$.

The tree distributions are the mixture components, and the λ_k are called mixture coefficients. A mixture of trees can be viewed as containing an unobserved choice variable z , which takes values $k \in \{1, \dots, m\}$ with probability λ_k . Conditioned on the value of z the distribution of the visible variables V is a tree. The m trees may have different structures and different parameters.

3 THE LEARNING ALGORITHM IN THE CONTEXT OF THE MT-FDA

We present first an algorithm for fitting a mixture of trees to an observed data set in the Maximum Likelihood paradigm via the Expectation-Maximization (EM) algorithm (Dempster, Laird, & Rubin, 1977). The mixtures of trees learning algorithm constitutes a building block for the FDA presented here, this algorithm can be found in (Meila, 1999) where it was introduced. We resume some of its features:

The learning problem is: Given a set of observations $D = \{x^1, x^2, \dots, x^N\}$, we are required to find the mixture of trees Q that satisfies

$$Q = \arg \max_Q \sum_{i=1}^N \log Q(x^i) \quad (3)$$

The EM algorithm introduces a likelihood function called the *complete log-likelihood* which is the log-likelihood of both, the observed and the unobserved data, given the current model estimate $M = \{m, T^k, \lambda_k, k = 1, \dots, m\}$

$$lc(x^{1 \dots N}, z^{1 \dots N} | M) = \sum_{i=1}^N \sum_{k=1}^m \delta_{k,z^i} (\log \lambda_k + \log T^k(x^i)) \quad (4)$$

where δ_{k,z^i} is equal to one if z^i is equal to the k th value of the choice variable, and zero otherwise.

The idea underlying the EM algorithm is to compute and optimize the expected value of lc . In the context of population based evolutionary optimization our objective is to find a factorization of the probability distribution of the selected set of points using a mixture of trees. We have called our algorithm Mixture of Trees FDA (MT-FDA).

In figure 1 the pseudo-code of the MT-FDA is presented. The first population is randomly generated by the algorithm. From the current population, a subset of points is selected. A mixture of trees Q that fits the selected set is found using the mixture of trees learning algorithm that takes as parameters the number of trees, and a schedule with the number of learning steps in each generation. New points are generated by sampling from Q . The best elitism scheme is used, where all the individuals selected in the current population pass to the next one. Different replacing strategies can be used to combine points from the current population and new generated points in the next population.

Mixture of trees FDA (MT – FDA)

- **STEP 0:** Set $t \leftarrow 0$. Generate $N \gg 0$ points randomly.
- **STEP 1:** Select a set S of $k < N$ points according to a selection method.
- **STEP 2:** Calculate a mixture of trees Q that approximates S using the mixture of trees learning algorithm.

- **STEP 3:** Generate $N - k$ new points sampling from Q .
- **STEP 4:** Combine in the new population the k selected points with the $N - k$ new points. Set $t \leftarrow t + 1$
- **STEP 5:** If the termination criteria are not met, go to STEP 1

Figure 1: MT-FDA

The determination of the extent of learning to be allowed is a sensitive issue for the MT-FDA. When sampling from the learned model, the two traditional goals of an efficient search, exploitation and exploration, have to be accomplished. A model that best approximates the data can less likely generate, during the sampling step, points that belong to unexplored areas of the search space. So, it is advisable to stop the learning algorithm before the improvement in the likelihood ceases. The number of trees also influence the likelihood. The design of strategies that set the appropriate schedule for the learning steps, and of criteria to determine the number of trees for a given optimization problem, are topics where further research is required.

We test two methods for selecting a convenient starting mixture of trees. Both methods begin by initializing each component of the mixture using the Chow and Liu's algorithm. They differ in the second step. The first method makes perturbations to the structure of each tree. The parent of one of the leaf nodes is replaced by another randomly selected node. By making this change on a leaf node the procedures guarantees no cycle will be formed in the tree after the perturbation.

The second method keeps intact the tree structure and makes a perturbation to the initial probability values $P^k(x^i)$, in order to let the learning algorithm change the structure when fitting the perturbed probabilities. Recall that, independently of the selected root, Chow and Liu's algorithm guarantees that all trees are equivalent in their representation of the data. By applying a small perturbation on $P^k(x^i)$ values, identicalness among the trees is broken. Both methods guarantee solutions superior to random initialization.

Mixture distributions have a number of distinctive attributes that make them particularly appealing for their use in the framework of FDAs. Maybe the most important is the possibility of representing, condensed in just one model, different patterns of interactions among the variables of the problem. In Bayesian Networks the change in one variable's value can determine changes only in the parameters of other variables, not

in their structural relation. In mixture of finite distributions the structure of dependencies among a set of variables can change depending on the values of the choice variable they depend on. This fact can be used in a flexible way for incorporating diverse search strategies into the MT-FDA.

Now we present some alternative ways the choice variable can be incorporated by an FDA whose model is a mixture of trees. The analysis is divided in two scenarios, when the choice variable is hidden, and when it is known.

3.1 The choice variable z is hidden.

Two cases can be distinguished in this scenario. The first is when, although the choice variable is unknown, it is in fact one (or a small subset) of the variables of the problem. In this case a fitter approximation of the selected points by means of a mixture of trees could be found by identifying the set of variables. There exist methods for identifying these variables from the analysis of data (Meila, 1999). Although we do not explore this trend in the present work, we hypothesize that this identification task can be inserted in the evolutionary process.

The second case, and the one extensively considered in our experiments, is when there is no previous information about the choice variable, but we try to fit the target distribution with a mixture that has m trees, based on an unknown variable that could help to cluster the space of solutions. As the existence of a mixture of trees based on such a kind of choice variable is an assumption, there could be cases where the mixture of trees does not lead to an appropriate factorization.

3.2 The choice variable z is known

Again we analyze two particular cases: When the choice variable is in fact a variable, (or the subset of the variables) of the problem. And, when the choice variable z does not belong to the set of variables of the problem but we can decide its values based on a predefined criterium. We mention two cases:

- a) z is related to a genotype clustering of the set of points
- b) z is related to a phenotype clustering of the set of points.

It has been shown (Pelikan & Goldberg, 2000) that for certain type of problems a genotype clustering of the population can contribute to an efficient search. The mixture of trees model can be used not only to cluster the space of solutions but also to exchange informa-

tion among the clusters during the evolution. When solutions could be classified in regions based on their structural similarities, and a mapping between solutions that belong to the same class and their fitness evaluation exists, it is expected that a mixture of trees based on a choice variable describing the classes could lead to a good approximation.

4 EXPERIMENTS

The experiments were designed to illustrate the behavior of the proposed algorithm and compare its performance with other FDAs. First we introduce the functions that were employed. Then we present a comparison between the MT-FDA and a tree based FDA. Some experiments are presented to illustrate the difference in the performance of the Bayesian FDAs and the MT-FDA. Finally an example is shown on the convenience of using one of the variables of the problem as the choice variable.

Let u be the number of bits turned on in the string x . A function of unitation is a function whose value depends only on the number of ones on an input string. The values of the strings with the same number of ones are equal. Deceptive functions are defined as a sum of more elementary deceptive functions f_k of k variables.

$$f(x) = \sum_{j=1}^l f_k(s_j), \quad (5)$$

where s_j are non-overlapping substrings of x containing k elements.

Function OneMax:

$$OneMax(x) = \sum_{i=1}^n x_i \quad (6)$$

Function f_{dec}^3 :

$$f_{dec}^3 = \begin{cases} 0.9 & \text{for } u = 0 \\ 0.8 & \text{for } u = 1 \\ 0.0 & \text{for } u = 2 \\ 1.0 & \text{for } u = 3 \end{cases} \quad (7)$$

Function $f_{3deceptive}$:

$$f_{3deceptive}(X) = \sum_{i=1}^{\frac{n}{3}} f_{dec}^3(X_{3i-2}, X_{3i-1}, X_{3i}) \quad (8)$$

Function *Isotorus*:

u	0	1	2	3	4	5
$Isot_1$	m	0	0	0	0	$m-1$
$Isot_2$	0	0	0	0	0	m^2

(9)

$$F_{Isotorus} = \quad (10)$$

$$\sum_{i=2}^n Isot_2(x_{up}, x_{left}, x_i, x_{right}, x_{down}) \quad (11)$$

$$+ Isot_1(x_{1-m+n}, x_{1-m+n}, x_1, x_2, x_{1+m}) \quad (12)$$

where x_{up} , etc., are defined as the appropriate neighbors, wrapping around.

Function *BigJump*:

$$BigJump = \begin{cases} u & \text{for } 0 \leq u \leq n-m \\ 0 & \text{for } n-m \leq u \leq m \\ k \cdot n & \text{for } u = m \end{cases} \quad (13)$$

Function Nf_{dec}^3 :

$$Nf_{dec}^3 = \begin{cases} \sum_{i=1}^{\frac{n-1}{3}} f_{dec}^3(X_{3i-1}, X_{3i}, X_{3i+1}), & \text{if } x_1 = 1 \\ \sum_{i=1}^{\frac{n-1}{3}} (1 - f_{dec}^3(X_{3i-1}, X_{3i}, X_{3i+1})), & \text{oth.} \end{cases} \quad (14)$$

4.1 Comparison between a tree based FDA and the MT-FDA

First we make a comparison between a tree based FDA and the MT-FDA. The tree based FDA has the same pseudo-code as the MT-FDA presented in section 4, but only one tree found using the Chow and Liu's algorithm is employed. The main difference between Baluja's algorithm and our tree based FDA is that we do not update the bivariate probabilities multiplying by a decay factor, instead in every generation, bivariate probabilities are calculated from the selected set. We study the difference in the behavior of the tree based FDA and the MT-FDA for diverse truncation values and functions.

Common parameters for the experiments were: Population size = 1000, Best elitism, Maxgen = 20. Stopping criteria were: the maximum number of generations, and a total homogeneity in the selected population (i.e. all the individuals were identical).

Function	n	τ	Tree		MT-FDA	
			S	Gen.	S	Gen.
	30	0.05	37	4.92	9	6.33
$f_{3deceptive}$	30	0.1	60	5.73	63	6.66
	30	0.15	72	6.84	75	7.8
	30	0.2	65	8.28	75	9.28
	30	0.25	85	8.8	95	9.56
<i>Isoturus</i>	36	0.05	80	3.49	75	4.05
	36	0.1	84	4.4	93	4.9
	36	0.15	85	5.19	92	5.78
	36	0.2	84	6.14	90	6.5
	36	0.25	71	6.86	91	7.38

Table 1: Numerical results for MT-FDA and tree based FDA

The schedule for learning was the following: In the first population there were 10 learning steps to learn the mixture, in the following generations 5 learning steps. Recall that for the MT-FDA the trees from which the learning algorithm is started are also found using the Chow and Liu’s algorithm. So this initialization can be seen as the best we could achieve with just one tree (what the tree based FDA actually does), but in the following learning steps the likelihood of data given by the initial trees is improved.

In Table 1 are presented results for two different functions. n is the number of variables, τ the truncation parameter, S the number of times the optimum was found in 100 runs and $Gen.$ the average number of generations needed to find the optimum. Initial experiments were made for $f_{3deceptive}$ with $\tau = 0.05$. Results were not particularly heart warming. In the table it can be appreciated that for this truncation value the tree based FDA finds the optimum four more times than when MT-FDA is used. However, for, $\tau = 0.25$ the MT-FDA clearly surpasses the tree with an impressive 95 percent of success. Similar results were achieved for the *Isoturus* function.

4.2 Comparison between Bayesian FDAs and the MT-FDA

In the general class of BNs the conditional probability of a variable x can depend on a subset of variables and not on only one like in the subclass of trees. All these variables are called parents. The complexity of the network is related to the maximum number of parents any variable x can have. BNs learning algorithms allow to incorporate constraints related with a maximum number of parents or the network complexity.

In (Etzeberria & Larrañaga, 1999) BNs were intro-

duced in the framework of Evolutionary Optimization to learn, in every generation, a factorization of the selected points. In this paper we compare our algorithm with other two Bayesian FDAs. The Bayesian Optimization Algorithm (BOA) introduced in (Pelikan, Goldberg, & Cantú-Paz, 1999) and the Learning Factorized Distribution Algorithm (LFDA) (Mühlenbein & Mahnig, 1999b). Both algorithms use a Bayesian metric to measure the goodness of every Bayesian structure found, and a search procedure to search in the space of possible structures. In the BOA it is possible to set the maximum number of parents in the learnt network. For the LFDA we will refer to results appeared in (Mühlenbein & Mahnig, 1999a). Contrary to the BOA, the LFDA uses the BIC score to find the Bayesian structure and it uses a parameter α that allows to control the network complexity.

Function	n	FDA	α / Trees	S
<i>OneMax</i>	30	LFDA	0.75	80
	30	LFDA	0.5	38
	30	LFDA	0.25	2
	30	Tree		89
	30	MT-FDA	2	77
	30	MT-FDA	5	19
	30	MT-FDA	10	0
<i>BigJump</i> (30,3,1)	30	LFDA	0.75	100
	30	LFDA	0.5	96
	30	LFDA	0.25	58
	30	Tree		99
	30	MT-FDA	2	97
	30	MT-FDA	4	91
	30	MT-FDA	6	77

Table 2: Numerical results for MT-FDA and the LFDA

A number of experiments were conducted to compare the behavior of the MT-FDA with the LFDA. In this case we used an elitism parameter of 1 (i.e. only the best individual is passed to the next population), truncation selection and Maxgen=20. Stopping criteria for MT-FDA were the same that in the previous experiments. In Table 2 column 4 refers to the number of trees and the network density for the MT-FDA and the LFDA respectively. Later, it is explained how these values were chosen.

A number of interesting features can be noticed from the analysis of the results shown in Table 2. For these functions the number of trees in the MT-FDA and the density of the BN in the LFDA play a similar role. This analogy is evident for function *BigJump*, when the number of trees or the density of the BN are in-

creased results are poorer. In the table, values used for the number of trees were selected trying to emulate the values for the density of the BN used in experiments published in (Mühlenbein & Mahnig, 2000), but we do not claim we have attained a perfect correspondence between the number of trees shown in Table 2 and the α values shown for the same functions. The analogy is related only to the progression of values for both parameters.

Results for function *Bigjump* are an example too that when structural learning is done, simpler models can be better than more complex ones. Regarding differences between the MT-FDA and the LFDA, and beyond the analogy between the number of trees and the network density, results were very similar.

It has been acknowledged that Bayesian FDAs are very sensitive to the selection method used (Pelikan, Goldberg, & Sastry, 2000). Thus, when we claim here that one algorithm is superior to the other for the analyzed functions, the assertion is only valid for the particular selection strategy employed. We have specified the selection methods and parameters used in every comparison between Bayesian FDAs and the MT-FDA.

4.3 Comparison between BOA and the MT-FDA

We also compare our algorithm with the Bayesian Optimization Algorithm (BOA) introduced in (Pelikan, Goldberg, & Cantú-Paz, 1999). BOA uses a Bayesian metric to measure the goodness of every Bayesian structure found, and a search procedure to search in the space of possible structures. The implementation of BOA used in this paper rests on the BOA platform available at (Pelikan, Goldberg, & Sastry, 2000). It employs the Bayesian Dirichlet equivalent (BDe) metric, a greedy search procedure, and decision graphs for making the search more efficient.

Table 3 shows the experimental results of the comparison between the BOA and the MT-FDA for 2 functions. The parameters of the algorithm were: Population size = 600, Tournament selection with tournament size = 4, the worst 85 percent of the current population is replaced by the new generated individuals. The BOA was set to stop when the optimum was found, or an early convergence occurred (univariate marginals higher than 0.95). The stop criteria for MT-FDA were the same as in previous experiments, but note that in general this setting is very different from the one used in the previous experiments.

In the table the fourth column represents the maximum number of parents and the number of trees

Function	n	FDA	P/ T	S	Gen.
<i>Isoturus</i>	36	BOA	2	86	7.29
	36	BOA	4	81	7.23
	36	BOA	6	78	7.4
	36	Tree		72	7.96
	36	MT	2	87	8.22
	30	MT	4	89	7.83
	36	MT	6	93	7.38
<i>f_{3deceptive}</i>	30	BOA	2	78	7.47
	30	BOA	4	77	8.12
	30	BOA	6	77	8.49
	30	Tree		37	10.46
	30	MT	2	54	11.26
	30	MT	4	60	11.3
	30	MT	6	69	10.87

Table 3: Numerical results for MT-FDA and the BOA

for the BOA and the MT-FDA respectively. Column five represents the number of times the algorithm has converged in 100 runs. In the experiments for the *f_{3deceptive}* function BOA clearly overperformed the MT-FDA. For this function two issues are worth to point out. The poor performance of the tree based FDA, and the observation that when the number of trees is increased the gap in the behavior between the MT-FDA and the BOA is reduced. Nevertheless, by further increasing the number of trees the MT-FDA does not achieve better results than BOA.

This situation is completely reversed for function *Isoturus*. For this function, as the number of trees is increased MT-FDA overperforms BOA, when the number of trees is 6 this difference is evident. The general conclusion from these experiments is that for certain kind of functions MT-FDA can overperform Bayesian-FDAs. One question remains open: which are the criteria that allow to decide whether it is more convenient to apply one FDA or another? We hypothesize MT-FDA can be more suitable for functions with general and scattered overlapping between the variables, that could be "covered" by a set of relatively dependent probabilistic models. This seems to be the case for the *Isoturus* function.

4.4 The role of the choice variable in the MT-FDA.

As it was briefly discussed in previous sections the use of the choice variable by the MT-FDA allows different and flexible ways of conducting the search. We conducted experiments to evaluate the convenience of using one of the variables of the problem as the choice

Function	n	N	z	T	S	Gen.
Nf_{dec}^3	31	2500	unknown	0.15	24	8.96
	46	3000	unknown	0.15	21	11.57
	61	3000	unknown	0.25	12	19.8
	31	1500	x_1	0.15	30	3.1
	46	1500	x_1	0.15	28	6.25
	61	2500	x_1	0.25	23	10.48

Table 4: Numerical results for the Nf_{dec}^3 function.

variable. These experiments are very preliminary but help to illustrate how the parameters of the mixture model can be conveniently used in the FDA framework. Table 4 shows the results for function Nf_{dec}^3 . This function has many global optimal that are triggered by variable x_1 . When $x_1 = 1$ there is just one optimum (all variables are set to 1), when $x_1 = 0$ there are $3^{\frac{n}{2}}$ optima (all possible combinations of 2 ones in every partition). For every setting 30 experiments were run using truncation selection and a maximum of 25 generations.

In the table it is shown a comparison between two MT-FDA with mixtures models composed of two trees. The mixture model of the first MT-FDA uses x_1 as its choice variable, for the second the choice variable is unknown as was the case in the previous experiments. When x_1 is treated as the choice variable the individuals that fulfill ($x_1 = 0$) are approximated by one tree, and those that satisfy ($x_1 = 1$) by the other. In this case there is not learning of the tree structures, only the mixture coefficients are calculated as $\lambda_0 = \frac{N_0}{N}$, $\lambda_1 = \frac{N_1}{N}$, where N_0 and N_1 are respectively the number of individuals that satisfy ($x_1 = 0$) and ($x_1 = 1$). Nevertheless such a mixture allows the MT-FDA to focus on the region of the space of solutions defined by the x_1 value. Eventually, as evolution advances, one of the coefficients becomes 0, and a tree based FDA is run from then on.

In Table 4 the improvements achieved by considering x_1 as the choice variable can be seen. This is a simple example of how the choice variable can be incorporated also as a tool for influencing the exploration and exploitation purposes of the search.

5 CONCLUSIONS AND FURTHER WORK

In this paper we have introduced a FDA based on mixture distributions. The MT-FDA is different to other simple connected based FDAs, and to Bayesian FDAs too. Contrary to Bayesian FDAs the MT model allows

that the structure of dependencies among a set of variables changes. Compared to simple connected FDAs, in the MT-FDA, dependencies between the parents of a given node can be represented.

The algorithm exhibits a number of characteristics that places it far apart from Gallagher's algorithm (Gallagher, Frea, & Downs, 1999). It is appropriate for problems with integer representation that can have many variables. The performance of Gallagher's algorithm is more sensitive to the number of variables. The type of mixture model both algorithms employ is different, the adaptive mixture model of Priebe (Priebe, 1994) is used in Gallagher's algorithm.

Compared to the Estimation of Mixture of Distribution Algorithm (EMDA) introduced in (Peña, Lozano, & Larrañaga, 2001) the algorithm exhibits also several differences. MT-FDA is not focused on the optimization of multiobjective functions although it can be employed in multiobjective optimization too. Another difference is that our mixture model allows components with different structures. The version of the EM algorithm we use in this paper is different to the one presented in (Peña, Lozano, & Larrañaga, 2001).

In this paper we have shown that the MT-FDA can overperform the tree based FDA for the functions considered. The algorithm is also better than Bayesian FDAs for some functions like Isoturus. Another achievement of this paper is to have presented different ways to influence the search by manipulating parameters related to the mixture of trees.

The computational complexity of the MT-FDA is mainly given by the Estimation and Maximization steps of the (EM) algorithm. In each iteration of the mixture of trees learning algorithm the running time of the estimation step is $\Theta(mnN)$, and for the maximization step is $\Theta(mnr_{\max}^2)$ where $r_{\max}$ is the maximum cardinality of the variables. Both steps are computationally expensive. We have not treated this question in the present work. Currently we are investigating how to reduce the computational burden associated to the EM algorithm in the context of the MT-FDA. Promising lines of research and further work are:

1. To design efficient learning schedules that help to diminish the number of evaluations.
2. To extend the use of mixtures to deal with more complex probabilistic models.
3. The use of MT-FDA as a method to implement parallel and distributed population based search.

Acknowledgments

We are grateful for constructive comments of anonymous reviewers. This work was funded by the Cuban Ministry of Science, Technology and Environment, under the project Low Cost Evolutionary Algorithms.

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